SCIENTIFIC ENVIRONMENTS

Scientists work in different environments depending on what they are doing. Whether in a classroom or a laboratory, they must work safely, responsibly, and sustainably.

1. LAB vs CLASSROOM

LABORATORY	CLASSROOM
Used for practical experiments and investigations.	Used for learning, discussions, and planning.
Has specialised scientific equipment and materials.	Has textbooks, notebooks, and learning resources.
Students perform experiments and collect data.	Students learn scientific concepts and skills.
Strict safety rules must be followed.	General classroom rules must be followed.
Students wear safety equipment when required.	Safety equipment is usually not needed.

Laboratory = Learning by doing

Classroom = Learning by understanding

2. LAB SAFETY BASICS

Safety is everyone's responsibility.

Before the Experiment

✔ Read all instructions carefully.

✔ Listen to your teacher.

✔ Tie back long hair.

✔ Wear safety goggles and a lab coat if required.

During the Experiment

✔ Follow instructions carefully.

✔ Handle equipment gently.

✔ Use chemicals only as directed.

✔ Keep your work area neat.

✔ Never taste or smell chemicals unless instructed.

✔ Do not run or play in the laboratory.

✔ Report spills or accidents immediately.

After the Experiment

✔ Turn off equipment if instructed.

✔ Clean your workspace.

✔ Wash your hands.

✔ Return equipment to its proper place.

Laboratory Safety Symbols

Symbol	Meaning
🥽	Wear safety goggles

Wear protective
gloves

Fire hazard

Warning – Be careful

Hazardous material

Fire extinguisher

First aid

3. SUSTAINABLE SCIENTIFIC PRACTICES

Scientists work in ways that protect people and the environment.

1. Reduce Waste

• Use only the materials you need.

• Avoid wasting paper and chemicals.

2. Reuse Materials

• Reuse equipment whenever possible.

• Use both sides of paper.

3. Recycle

• Separate recyclable materials.

• Dispose of waste correctly.

4. Save Water and Energy

• Turn off taps after use.

• Switch off lights and electrical equipment when not needed.

5. Use Eco-friendly Materials

• Choose safer, non-toxic materials whenever possible.

• Avoid unnecessary plastic waste.

6. Respect Nature

• Protect plants and animals.

• Keep the surroundings clean.

• Do not litter.

Sustainable Scientist Checklist

Use only what you need.

Reduce waste.

Reuse materials.

Recycle correctly.

Save water.

Save electricity.

Keep the laboratory clean.

Dispose of waste safely.

Real-Life Examples

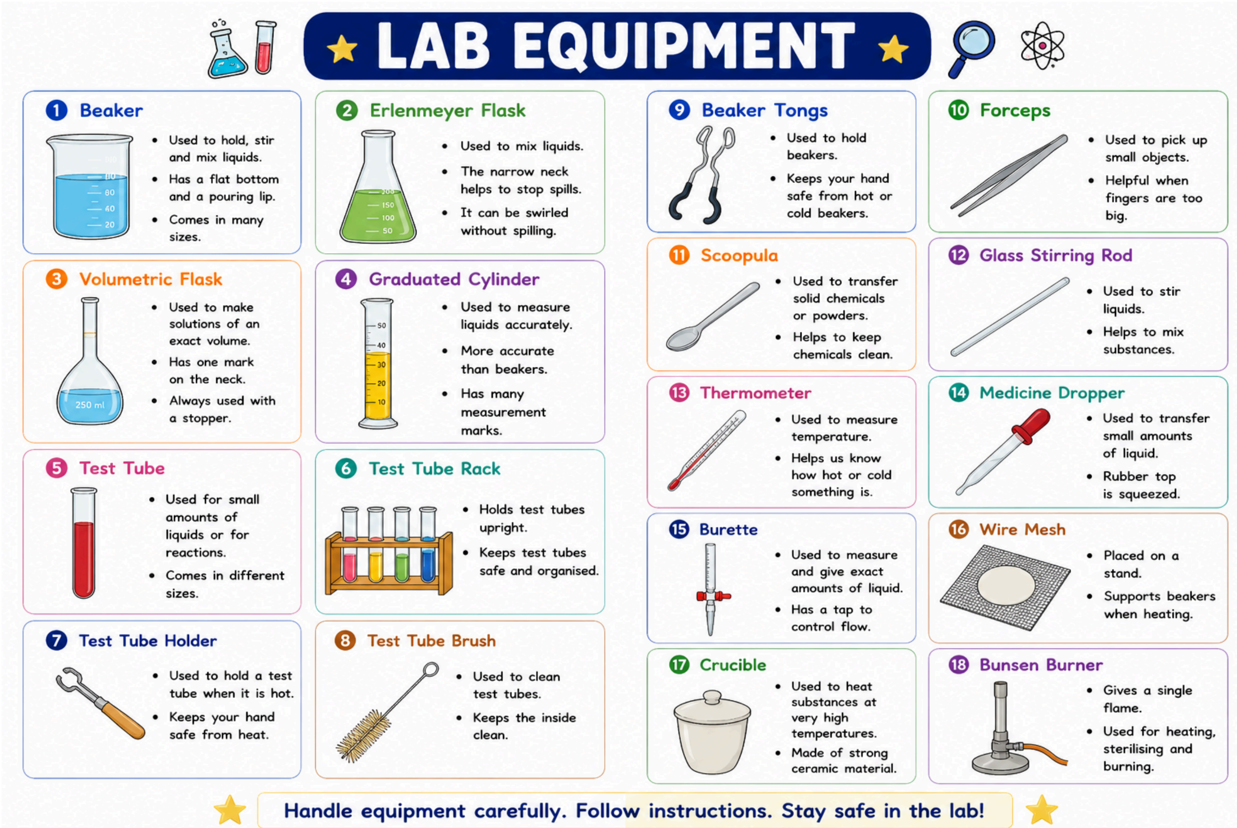
Situation	Scientific Practice

Using both sides of paper	Reduce paper waste
Turning off lights after an experiment	Save energy
Washing only the required equipment	Save water
Using reusable glass beakers instead of disposable cups	Reuse materials
Putting paper into the recycling bin	Recycle waste

Think Like a Scientist

Before starting an investigation, ask yourself:

- Am I working safely?
- Am I following instructions?
- Am I protecting the environment?
- Can I reduce waste?
- Can I reuse or recycle materials?


LAB EQUIPMENT
1 Beaker
80
60
40
20
Used to hold, stir and mix liquids.
Has a flat bottom and a pouring lip.
Comes in many sizes.
2 Erlenmeyer Flask
200
150
100
50
Used to mix liquids.
The narrow neck helps to stop spills.
It can be swirled without spilling.
3 Volumetric Flask
250 ml
Used to make solutions of an exact volume.
Has one mark on the neck.
Always used with a stopper.
4 Graduated Cylinder
50
40
30
20
10
Used to measure liquids accurately.
More accurate than beakers.
Has many measurement marks.
5 Test Tube
Used for small amounts of liquids or for reactions.
Comes in different sizes.
6 Test Tube Rack
Holds test tubes upright.
Keeps test tubes safe and organised.
7 Test Tube Holder
Used to hold a test tube when it is hot.
Keeps your hand safe from heat.
8 Test Tube Brush
Used to clean test tubes.
Keeps the inside clean.
9 Beaker Tongs
Used to hold beakers.
Keeps your hand safe from hot or cold beakers.
10 Forceps
Used to pick up small objects.
Helpful when fingers are too big.
11 Scoopula
Used to transfer solid chemicals or powders.
Helps to keep chemicals clean.
12 Glass Stirring Rod
Used to stir liquids.
Helps to mix substances.
13 Thermometer
Used to measure temperature.
Helps us know how hot or cold something is.
14 Medicine Dropper
Used to transfer small amounts of liquid.
Rubber top is squeezed.
15 Burette
Used to measure and give exact amounts of liquid.
Has a tap to control flow.
16 Wire Mesh
Placed on a stand.
Supports beakers when heating.
17 Crucible
Used to heat substances at very high temperatures.
Made of strong ceramic material.
18 Bunsen Burner
Gives a single flame.
Used for heating, sterilising and burning.
Handle equipment carefully. Follow instructions. Stay safe in the lab!